Image Denoising with Block Matching and 3D Filtering (BM3D)

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Introduction

- Additive White Gaussian Noise (AWGN) degrades image quality by introducing random variation in pixel luminance
- The Block Matching and 3D Filtering algorithm (BM3D) reduces AWGN by stacking similar pixel blocks into a 3D transform to collaboratively distinguish signal from noise
- Computationally cheaper than deep learning denoising while achieving similar results
- Connects classical image processing concepts in filtering and histogram interpretation with advanced collaborative filtering techniques

Step 1 - Block Matching

For each reference patch, x_R, in the image:

1. Extract small reference pixel patch (eg. 8×8)
2. Find all patches with distance less than a certain threshold
3. Distance equals squared pixel difference (Parseval's Theorem)

$$d(Z_{x_R}, Z_x) = \Sigma(Z_{x_R} - Z_x)^2$$

Eq. 1

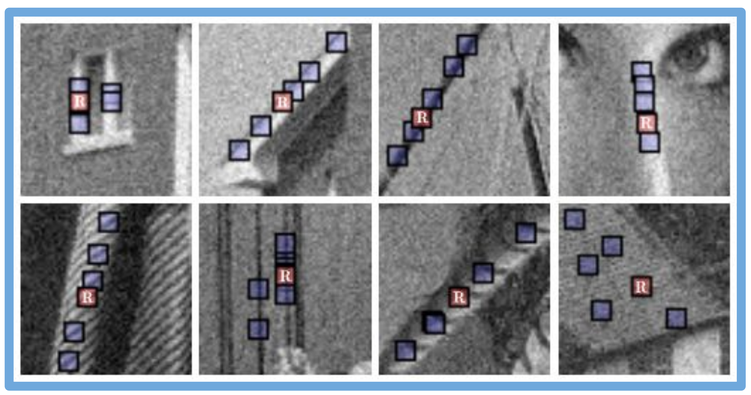

Figure 1. [block matching visualization 1]. From "Image denoising by sparse 3D transform-domain collaborative filtering" by Kostadin Dabov, Alessandro Foi, Vladimir Katkovnik, and Karen Egiazarian, Senior Member, IEEE, (2007), *IEEE TRANSACTIONS ON IMAGE PROCESSING, VOL. 16, NO. 8*, p. 4. https://webpages.tuni.fi/foi/GCF-BM3D/BM3D_TIP_2007.pdf.

Step 2 - 3D Collaborative Filtering

Apply 3D transform to the block group:

1. Pass 1 - hard-threshold transform coefficients zeros out sparse noise
2. Pass 2 - Wiener filter with signal power estimate weights coefficients by signal/noise ratio

$$\hat{Y}^{xR}_{xm} = \Gamma\, T_{3D} Z_{SxR}$$

Eq. 2

- Dark-boardered patches are references. Similar patches are stacked to form block
- Figure shows how similar patches are transformed for filtering.
- Simple background image to demonstrate block formation from patches

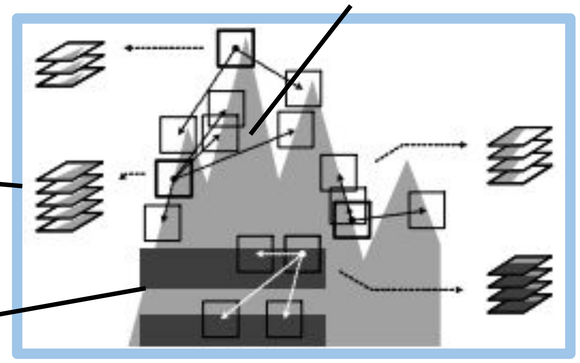

Figure 2. [block matching visualization 2]. From "Image denoising by sparse 3D transform-domain collaborative filtering" by Kostadin Dabov, Alessandro Foi, Vladimir Katkovnik, and Karen Egiazarian, Senior Member, IEEE, (2007), *IEEE TRANSACTIONS ON IMAGE PROCESSING, VOL. 16, NO. 8*, p. 3. https://webpages.tuni.fi/foi/GCF-BM3D/BM3D_TIP_2007.pdf.

Step 3 - Aggregation

Each pixel belongs to many overlapping groups.

1. Take weighted average of all group estimates (W)
2. pixels that appear in many high-quality groups get a more confident estimate
3. pixels that only appear in poorly matched groups are weighted less

$$\hat{y}(x) = \frac{\Sigma_{xR \in X} \Sigma_{xm \in Sxr} \hat{Y}^{xR}_{xm} \times W_{xR}}{\Sigma_{xR \in X} \Sigma_{xm \in Sxr} I(x \text{ is in patch around } x) \times W_{xR}}$$

Eq. 3

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Implementation

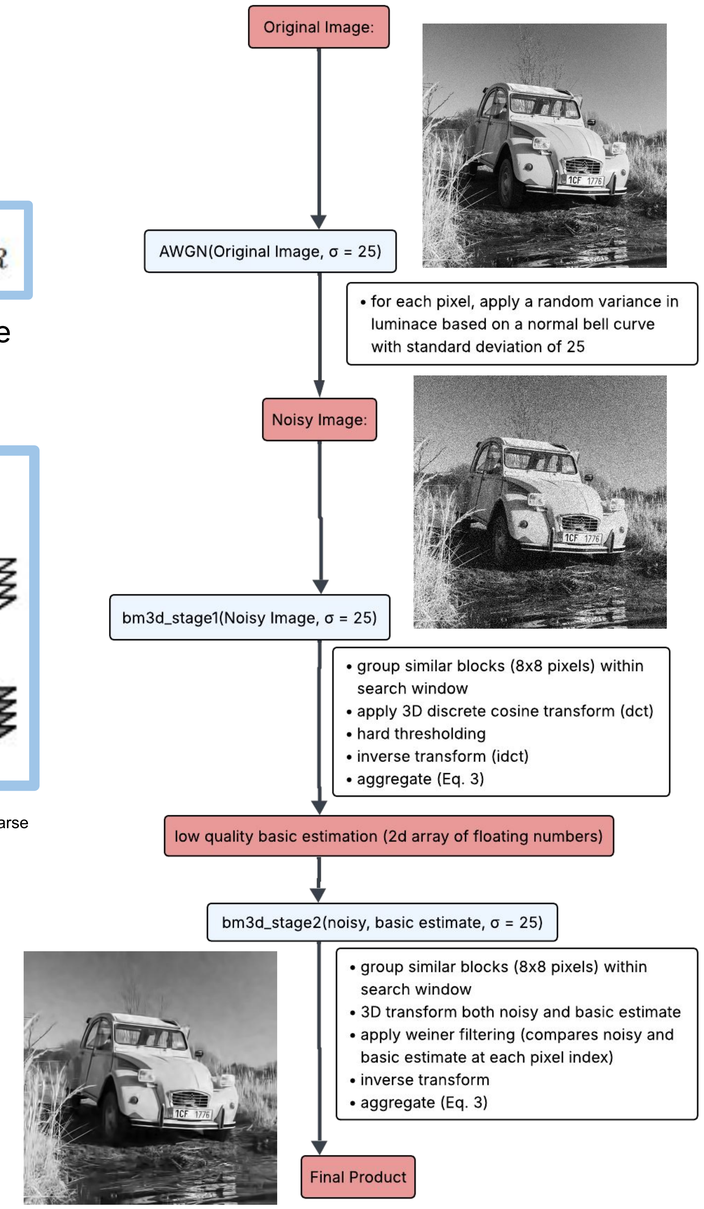

Figure 3. Full Pipline of Code Implementation

Original Image: **AWGN (σ = 25):** **Denoised:**

Figure 4. Histogram comparison:

Acknowledgments

Dabov, K., Foi, A., Katkovnik, V., & Egiazarian, K. (2007). Image denoising by sparse 3- D transform-domain collaborative filtering. IEEE Transactions on Image Processing, 16(8), 2080–2095.

Hambretch, F. (2013). 2.4 BM3D for Image Denoising — Image Analysis Class 2013. Retrieved from https://www.youtube.com/watch?v=BlDl6M0go-c